

MicroLA C++ Standard Feature Progression

«Rectangle»

C++20 (Baseline - Required)

Core C++20 Features

- Concepts:
 - Arithmetic<T>
 - FloatingPoint<T>
 - Integral<T>
- if constexpr / constexpr loops
- inline variables
- MICROLA_BIT_CAST (std::bit_cast)
- consteval (MICROLA_CONSTEVAL)

MicroLA Usage:

Type-safe templates
Branch optimization
Safe type punning
Compile-time validation
(stack-size config check)

Automatic Feature Detection:

C++20 is the required baseline.
Optional features prefer feature-test macros:
__cpp_concepts
__cpp_consteval
__cpp_if_consteval
__cpp_lib_bit_cast

C++26 constexpr math remains
guarded by active toolchain/library support.

Adds

«Rectangle»

C++23 Enhancements

MICROLA_IF_CONSTEVAL

- if consteval (MICROLA_IF_CONSTEVAL)

MicroLA Usage:

Dual compile/runtime paths
Optimized constexpr detection

Adds

«Rectangle»

C++26 (Latest)

MICROLA_CONSTEXPR_TRIG

- constexpr math functions:
 - std::sin, std::cos
 - std::sqrt, std::atan2

MicroLA Usage:

Compile-time rotations
Compile-time vector/quaternion norms
Compile-time Euler extraction
Arbitrary angle matrices
Full constexpr transformation

Example:

```
constexpr auto rot =  
    Mat<float,2,2>::rotation(45°);
```